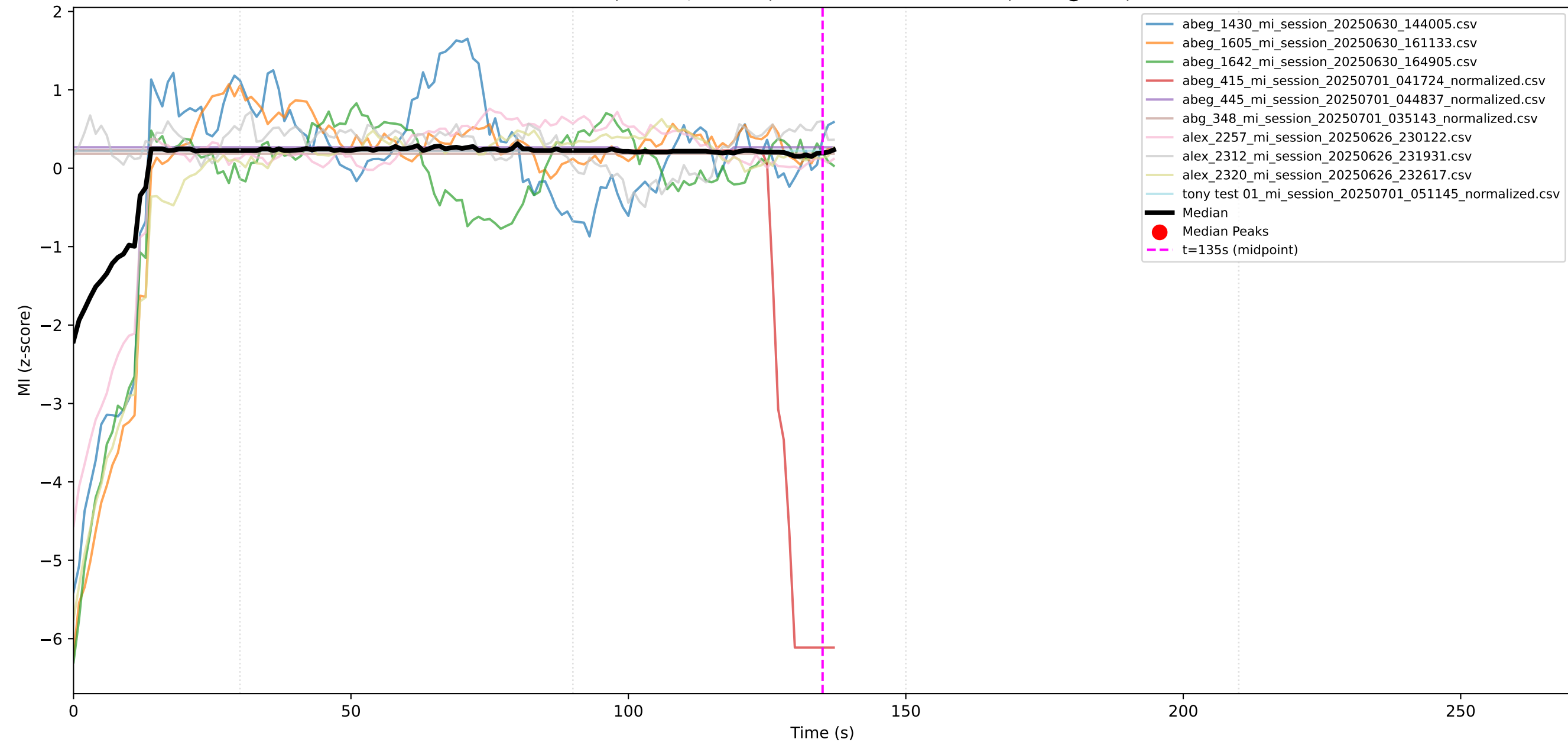
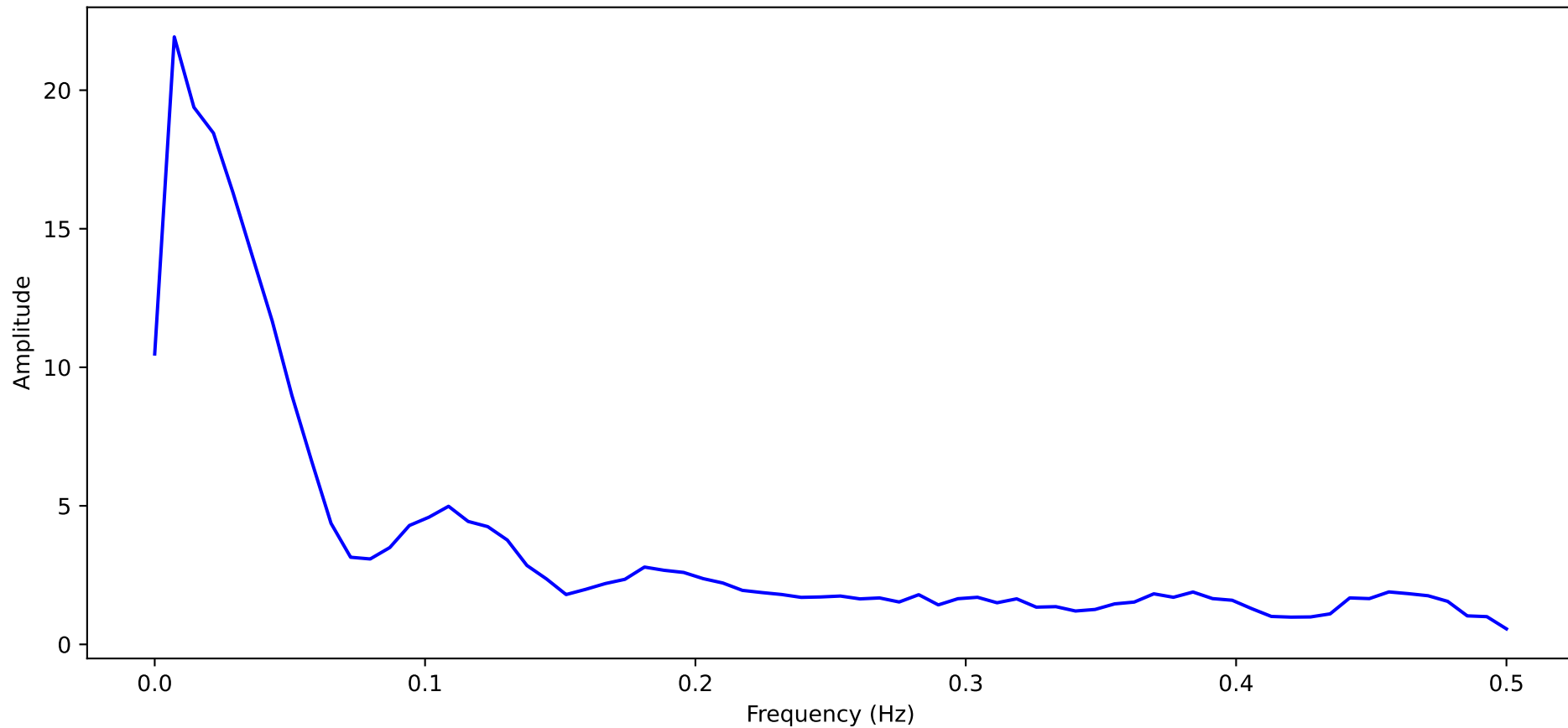


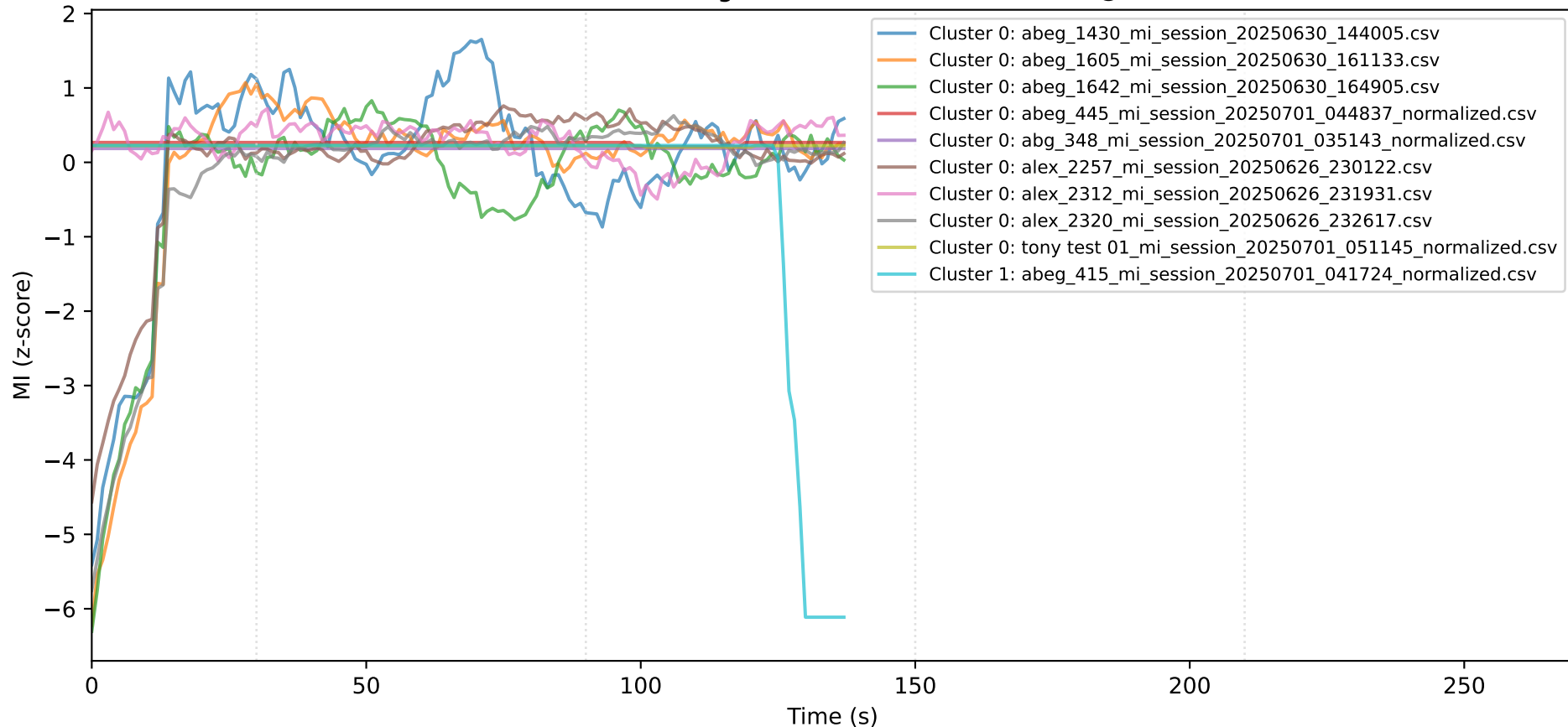
MI Curves (z-score, MA=20) with Median and Peaks (270s @ 1Hz)



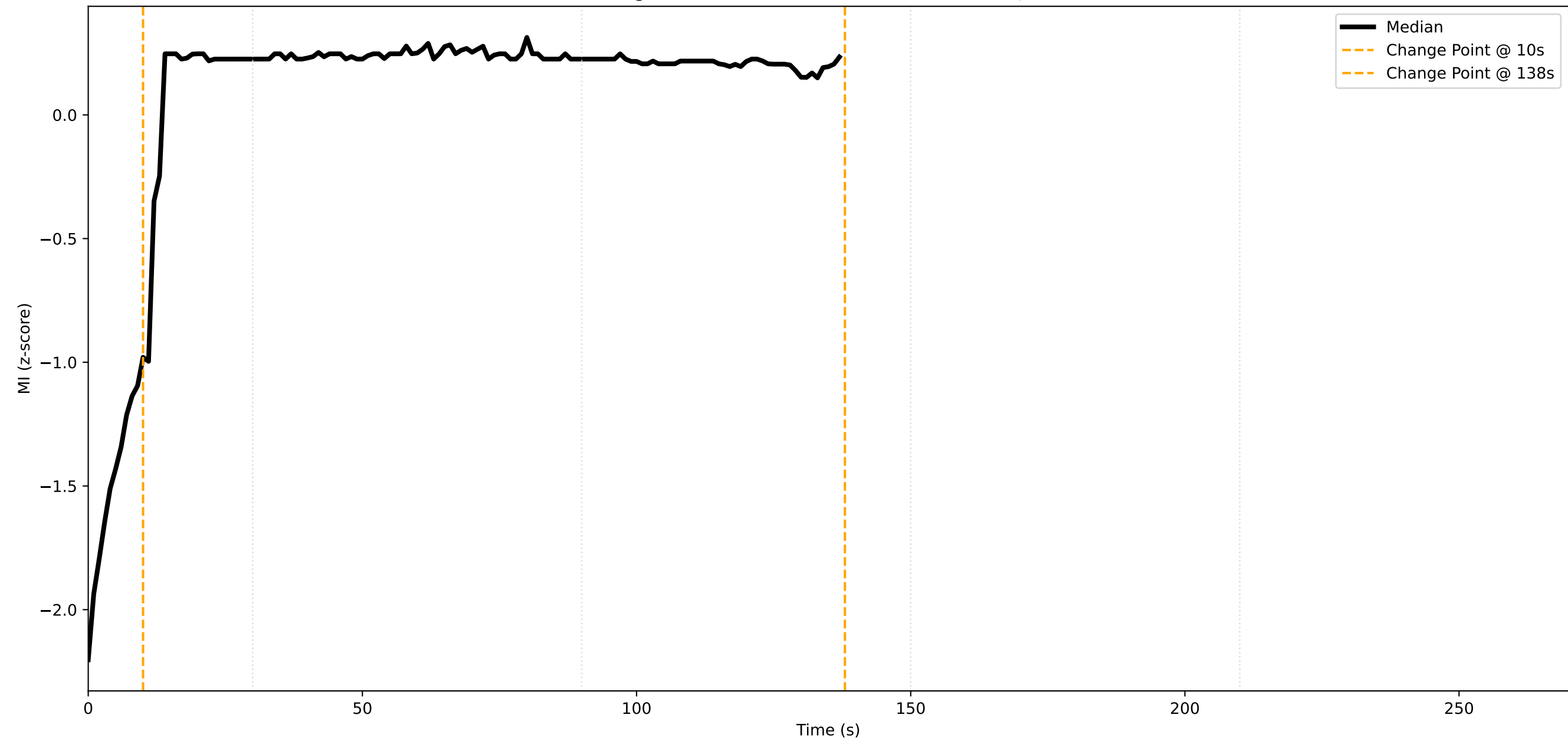
Median MI Frequency Spectrum (FFT)



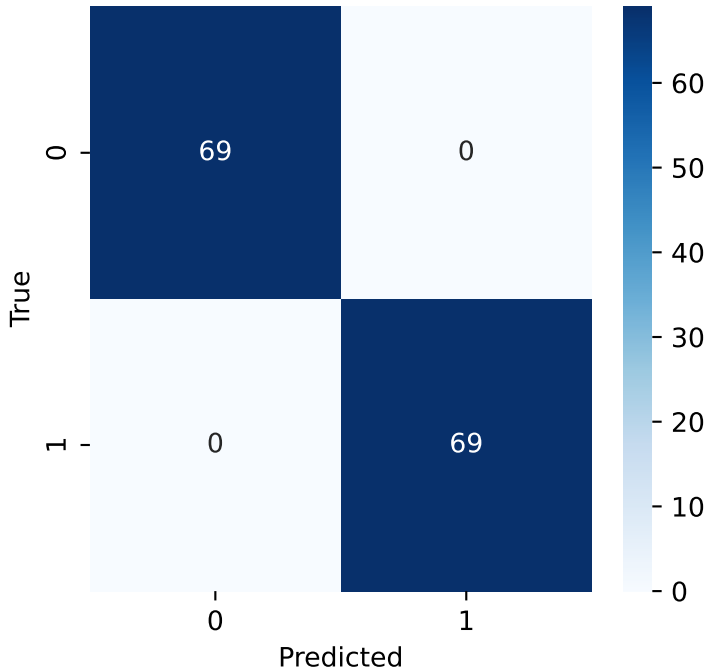
KMeans Clustering of MI Curves (k=2) - 270s @ 1Hz



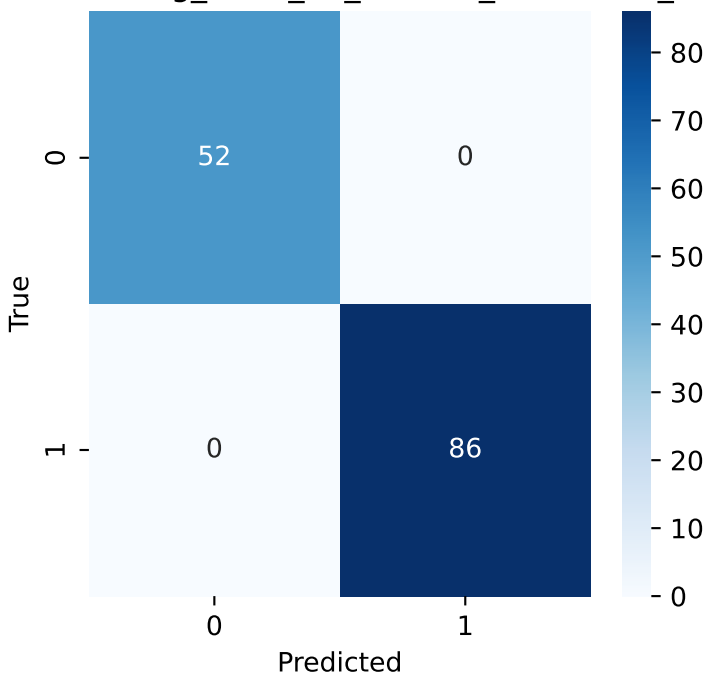
Change Point Detection on Median MI - 270s @ 1Hz



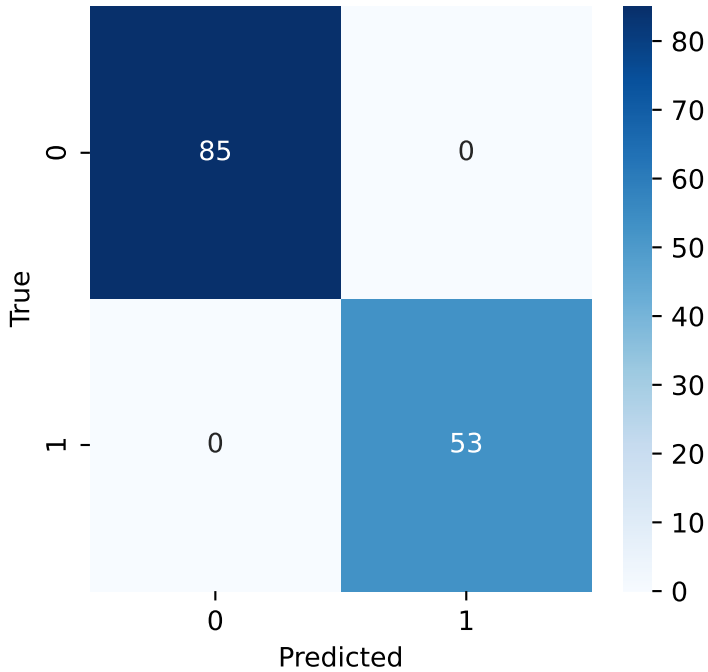
Matrix: abeg_1430_mi_session_20250630_1440



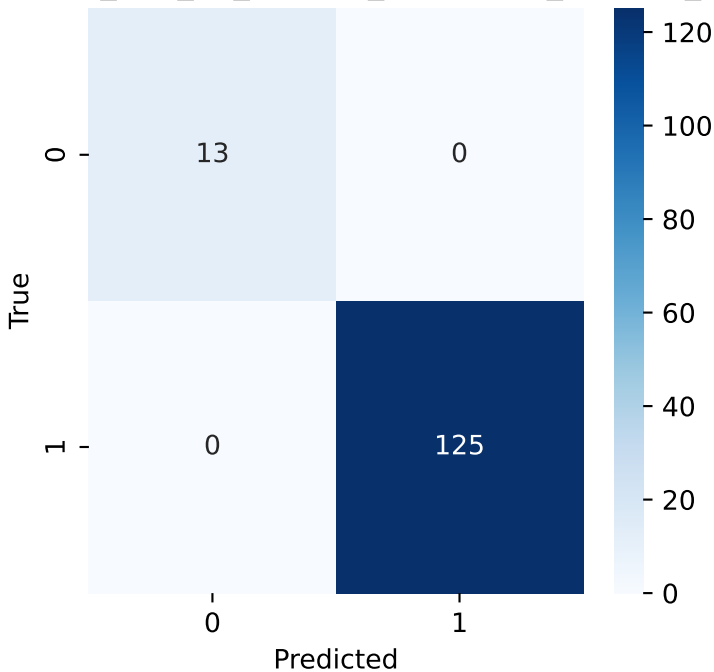
Matrix: abeg_1605_mi_session_20250630_1611



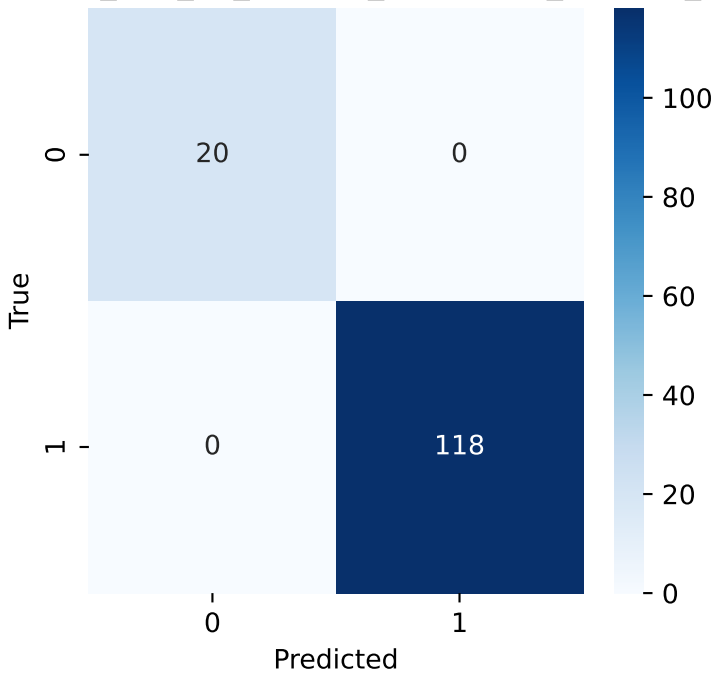
Matrix: abeg_1642_mi_session_20250630_1649



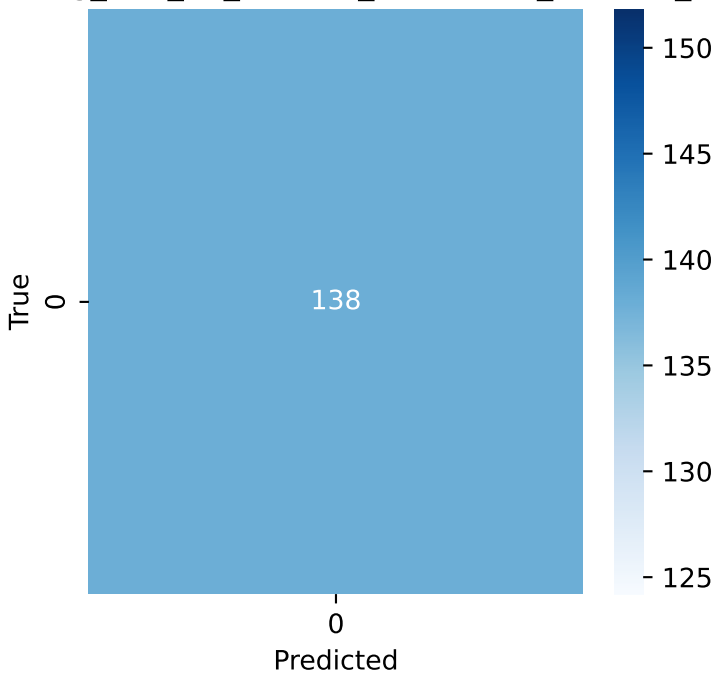
x: abeg_415_mi_session_20250701_041724_no



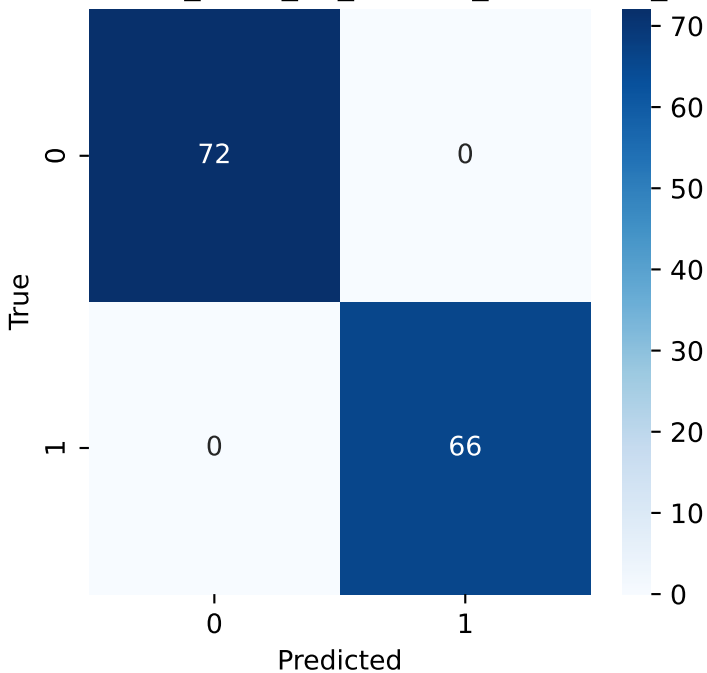
x: abeg_445_mi_session_20250701_044837_no



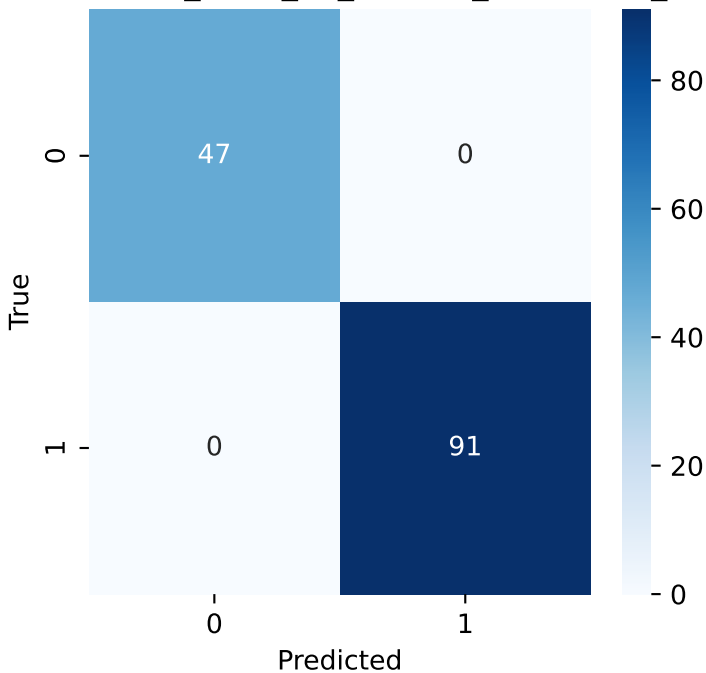
ix: abg_348_mi_session_20250701_035143_nor



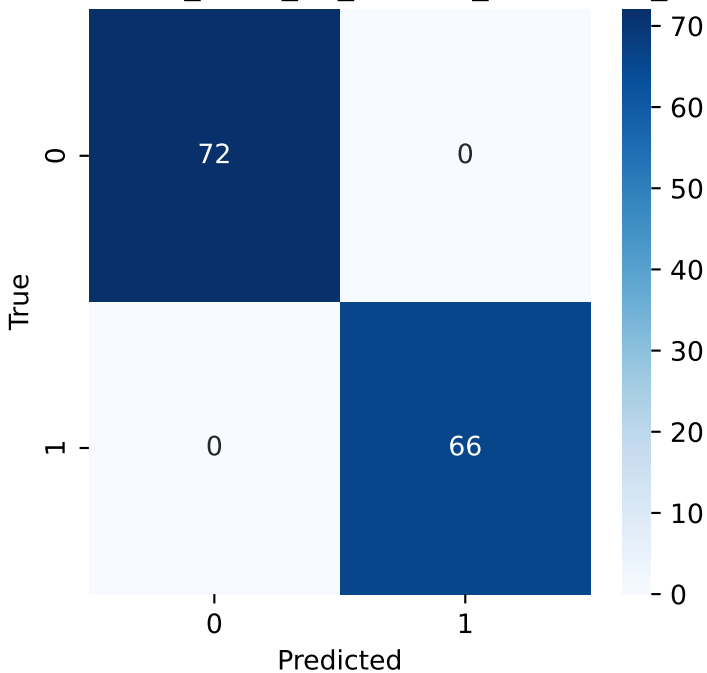
h Matrix: alex_2257_mi_session_20250626_2301



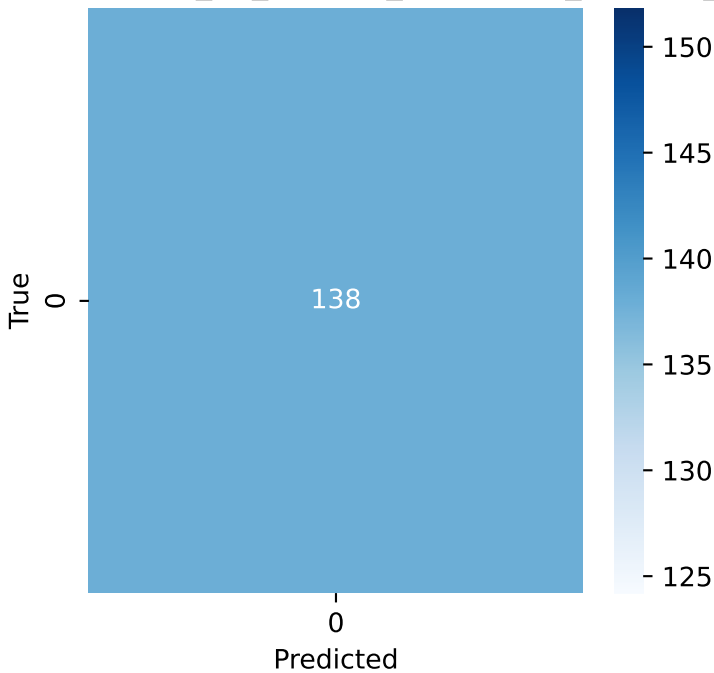
h Matrix: alex_2312_mi_session_20250626_2319



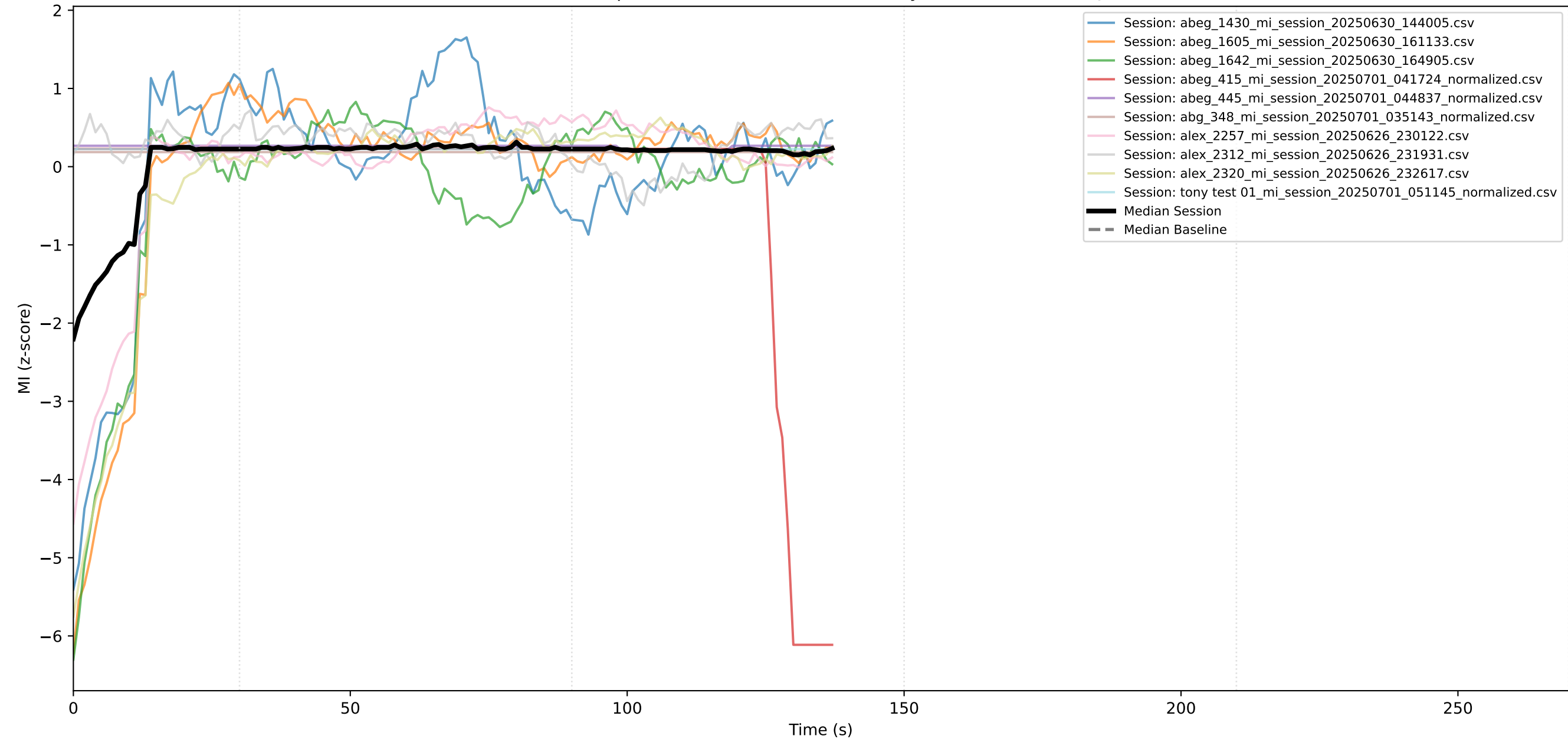
h Matrix: alex_2320_mi_session_20250626_2326



: tony test 01_mi_session_20250701_051145_no



Session MI (up to 10) vs. Median Baseline (Eyes Closed) - 270s @ 1Hz



MI Advanced Report

Generated: 20250717_023922

MI Advanced Report - Statistical Summary

1. Mindfulness Index (MI) Trajectory

- * All MI session files with ≥ 100 samples were included.
- * Individual and median MI curves were plotted (see Figure 8.1).
- * RM-ANOVA across six epochs (0-45s, 46-90s, 91-135s, 136-180s, 181-225s, 226-270s):
Anova

```
=====
F Value Num DF Den DF Pr > F
-----
Epoch    nan 5.0000 45.0000    nan
=====
```

- * Post-hoc pairwise comparisons (Tukey HSD):
Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
group1 group2 meandiff p-adj lower upper reject
-----
0-45s 136-180s -0.1178 nan nan nan False
0-45s 181-225s    nan nan nan nan False
0-45s 226-270s    nan nan nan nan False
0-45s 46-90s  0.5435 nan nan nan False
0-45s 91-135s  0.3571 nan nan nan False
136-180s 181-225s    nan nan nan nan False
136-180s 226-270s    nan nan nan nan False
136-180s 46-90s  0.6612 nan nan nan False
136-180s 91-135s  0.4749 nan nan nan False
181-225s 226-270s    nan nan nan nan False
181-225s 46-90s    nan nan nan nan False
181-225s 91-135s    nan nan nan nan False
226-270s 46-90s    nan nan nan nan False
226-270s 91-135s    nan nan nan nan False
46-90s 91-135s -0.1864 nan nan nan False
=====
```

- * Time-frequency analysis (FFT) of the median MI curve was performed (see Figure 8.3).

2. MI Dynamics

- * Peak analysis (median curve): 0 peaks ($z > 1.5$) at times [], amplitudes [], durations [] (see Table 8.2, Figure 8.2).
- * AUC (median): 11.46; mean $\pm$ SD (individual): 4.31 $\pm$ 25.87 (see Table 8.3).
- * Bootstrap 95% CI for mean MI difference (0-135s vs 136-270s): mean diff = 0.000, CI = [0.000, 0.000].
- * Change point detection (Pelt algorithm): 1 change points at [10] (see Figure 8.5).

3. Clustering & Mixed-Effects

- * K-Means clustering ($k=2$): cluster averages {}, user distribution {0: 9, 1: 1} (see Figure 8.6).
- * Mixed-effects model (MI ~ time + (1|Subject)): Mixed Linear Model Regression Results

```
=====
Model:          MixedLM Dependent Variable: value
No. Observations: 1380 Method:          REML
No. Groups:      10 Scale:              1.0048
Min. group size: 138 Log-Likelihood:    -1977.5717
Max. group size: 138 Converged:         Yes
Mean group size: 138.0
=====
Coef. Std.Err. z P>|z| [0.025 0.975]
-----
Intercept -0.307 0.081 -3.815 0.000 -0.465 -0.149
time       0.005 0.001 7.056 0.000 0.003 0.006
file Var   0.036 0.020
=====
```